



Live @Mars

DIYs are so fun, so is living on Mars. Here's how you can build your home on Mars.
<http://dgit.in/City-Mars>

Once in a decade!

Early to bed and early to rise, this Feb you will be able to see five planets with the naked eye, line up after a decade. <http://dgit.in/5LineUp>

standard achieve that higher data transfer rate. While none of the main organizations working on the upcoming standard haven't set a proper set of rules about what 5G should be, there is a wish list which is slowly gaining industry acceptance. These are some of the standards 5G is expected to meet:

1. 5G based technology is expected to handle a large amount data in tens of megabytes, catering to a group of thousands of people, without dropping or lowering network speeds. This would mean efficient and faster connectivity with speeds of upto 10Gbps.
2. It is expected to offer latency of less than 1 millisecond end-to-end round trip delay. This could mean a sublime VR experience over network. It is also important for mission critical tasks. for example, a doctor operating remotely on a patient via the help of a surgical robot.
3. Supreme bandwidth of a 1,000 times per unit area, which can support a hundred thousand simultaneous active connections. For this, 5G could harness or evolve upon the current and previous gen technologies such as 2G, 3G, 4G and Wi-Fi.
4. It is being perceived that 5G could bring 99.999% availability which could effectively mean the highest connection rates between the cell towers.
5. 5G is also suggested to cover 100% of the geographical area. This could essentially offer better connectivity over remote terrain and rural areas. If the upcoming standard is able to achieve this goal in real life scenarios, it will become the most important factor in the growth of developing nations.
6. One of the drawbacks with 4G is that it is not battery efficient, but 5G may change that. Research teams are working to reduce the power consumption by up to 90% at the terminal end, which will in turn make the whole process cost effective. The power consumption at a user's mobile level is also expected to go down which will make cellular devices last longer.

It is not guaranteed that these set requirements will be matched perfectly even by 2020, which is the expected year

we might have some concrete technologies to support the above mentioned requirements. Albeit, early tests and studies in the field suggest that we are close enough.

Fact

4G is likely to evolve and give us its transitional standards. 4.5G or as it is unofficially called LTE-Advanced Pro. In November 2015, HKT and Huawei successfully demoed the world's first 4.5G network offering 1Gbps of speed.

Why do we need 5G?

The answer to that is quite simple. 5G is vital to solve critical technological problems and keep up with future innovations. Another factor is the exponential growth in the user base, which is predicted to balloon beyond the seven billion mark by 2020. The current generation technology is not powerful or efficient enough to handle this massive growth, comprising mostly of traffic generated by video consumption these days.

Beyond that, 5G could be the key to seamless integration of technology within our daily lifestyle, in a more automated world, where we strive to achieve by the end of this decade. Automated vehicles, IoT devices, 4K and 8K streaming, fast and responsive health care and emergency services, are just some of 5G's countless use-cases. To accommodate all these varying needs, 5G is required. It's benefits are far stretched out than any of the previous generations. From connecting billions of people to offering faster internet speeds, it has to do it all and do it well.

How will it be different from 4G?

First, the speed. Yes, if boffins in the research lab are able to crack the mentioned wish list, we might get 10Gbps connection speeds by the end of 2020. This means a Full HD movie of around 8GB in size will take just six seconds to download instead of the 6-7 minutes it would take over normal 4G connections. 5G also brings the promise of better coverage, meaning lower call drop rates and more connected devices, currently a major problem in India. On top of that, 5G is expected to be quite flexible in terms of

its infrastructure, making it easier to be deployed under low-costs. If this becomes a fact, India will be able to connect at a more rapid pace than ever.

We can expect, that by 2020, technology will progress enough to make a bigger leap in remote computing and VR. It's also safe to presume that communication will become extremely real-time, offering a tactile internet, where you would be able to not only see and listen to it but touch and feel it as well. Also, IoT and connected devices such as wearables, smart home appliances and even connected vehicles will become a more intrinsic part of the



future. They will obviously require faster and more reliable networks to accommodate the surge in growth.

On a technical level, 5G is likely to use higher frequency bands which are currently being used by radar, satellites and military systems. This is the 30Ghz to 300GHz frequency band which is also called the millimeter-length wavelength. These waves are given consideration since they are open to huge amount of bandwidths and this spectrum is relatively less occupied compared to other spectrum bands in use. These can carry a large amount of data at higher speeds. However, millimeter waves are not perfect, their coverage is less than 300 metres, they don't penetrate walls and become woeful with changes in atmosphere such as rain.

Many researchers are working on newer technologies based on Wi-Fi and other waveforms such as FBMC (Filter